

REQUIREMENT ANALYSIS

Data Flow Diagrams (DFD) and User Stories

Date	04 November 2025
Team ID	NM2025TMID05906
Project Name	Garage Management System (GMS)
Maximum Marks	4 Marks

Data Flow Diagram (Level 0 – Context Diagram)

The Level 0 DFD shows the overall flow of information between key entities.

Entities:

1. Customer – Registers, submits service request, receives updates.
2. Mechanic – Views assigned jobs and updates status.
3. Admin – Assigns jobs and monitors operations.

Flow Summary:

1. Customer submits a service request.
2. System stores the request.
3. Admin assigns a mechanic.
4. Mechanic updates service status.
5. System notifies customer.

Data Flow Diagram (Level 1 – Detailed View)

Steps:

1. Customer logs in and submits service request.
2. Data stored in Service_Request object.
3. Admin assigns mechanic. 4. Mechanic receives notification.
5. Mechanic updates service status.
6. Updates sent to customer.
7. Inventory logs spare parts usage.
8. Admin closes service.
9. Reports auto-generated.

USER STORIES

User Story ID	Role	Description / Goal
US1 – Customer Registration and Login	Customer	As a customer, I want to register and log in so that I can request and track services.
US2 – Submit Service Request	Customer	As a customer, I want to submit vehicle issue details so the garage can process my request.
US3 – Receive Status Notifications	Customer	As a customer, I want to receive service status notifications so I stay informed.
US4 – Assign Mechanic	Admin	As an admin, I want to assign a mechanic to each service so it is handled efficiently.
US5 – View Assigned Jobs	Mechanic	As a mechanic, I want to view my assigned tasks so I can complete them.
US6 – Update Service Status	Mechanic	As a mechanic, I want to update job progress so the customer and admin are informed.
US7 – Manage Inventory	Admin	As an admin, I want to track spare parts usage so inventory stays accurate.
US8 – Generate Reports	Admin	As an admin, I want to generate service and performance reports so I can monitor operations.